31N0405

Roll No

Total No. of Pages.: 04

31N0405

B.Tech III Sem Main/Back Exam 2025-26

Computer Science and Engineering

(3CS4-05) Software Engineering

3AD4-05,3AM4-05,3AI4-05,3CM4-05,3CY4-05,3CD4-05,3CS4-05, 3IT4-05,3IC4-05 - Common to CS, IT, AI, DS,MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours

Maximum Marks: 70
Min. Passing Marks:

Instructions to Candidates:

Part-A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All ten questions are compulsory.

Part- B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1__NIL_____

2___NIL_____

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Part-A [2x10=20]

Q.1. Why is software requirements?

Q.2. What is the significance of software life-cycle models?

Q.3. How is verification different from validation?

Q.4. What is the role of a data dictionary in requirement analysis?

Q.5. What is the significance of Finite State Machines (FSM) Models in analyzing requirements?

Q.6. What are the fundamentals of software design?

Q.7. What are the difference between data architectural and procedural design.

Q.8. What are the key concepts of Object-Oriented Design (OOD)?

Q.9. What are the relationships between classes and objects in OOD?

Q.10. Discuss the characteristics of effective modular design?

PART-B [5x4=20]

Q1. Explain the difference between formal and informal requirements specification.

Q2. Explain the concepts of LOC (Lines of Code) and FP (Function Points) in software estimation.

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Q.3. Discuss the COCOMO estimation model and its significance in project management.

Q.4. Why is risk analysis an essential aspect of software project management?

Q.5 How does software prototyping contribute to the requirement analysis process?

Q.6 Discuss the characteristics of effective modular design.

Q.7. Discuss the concept of object modularization in Object-Oriented Design.

PART- C **[3x10=30]**

Q.1. Analyze the impact of not having a formal requirements specification on the overall software development process.

Q.2. Explain the assessment of risks associated with a specific software project and develop a risk mitigation strategy.

Q.3. Critically evaluate the benefits and drawbacks of using software prototyping in the requirement analysis phase.

Q.4. Critically assess the trade-offs between different design principles in a given software design scenario.

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Q.5.) Given a system, analyze the relationships between classes and objects to optimize the Object-Oriented Design.

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